

Sustech 2025南科大入学考试数学样卷

NO. 1

The equation of a curve is $y = (2k - 3)x^2 - kx - (k - 2)$, where k is a constant. The line $y = 3x - 4$ is a tangent to the curve.

Find the value of k . [5]

Mark scheme

Question	Answer	Marks	Guidance
6	$(2k - 3)x^2 - kx - (k - 2) = 3x - 4$	*M1	Equating curve
	$(2k - 3)x^2 - (k + 3)x - (k - 6) [=0]$	DM1	Forming a 3-term quadratic
	$(k + 3)^2 + 4(2k - 3)(k - 6) [=0]$	DM1	Use of discriminant (dependent on previous M mark)
	$9k^2 - 54k + 81 [=0]$ [leading to $k^2 - 6k + 9 = 0$]	M1	Simplifying and their 3-term quadratic in k
	$k = 3$	A1	
	Alternative method for Question 6		
	$(2k - 3)x^2 - kx - (k - 2) = 3x - 4$	*M1	Equating curve
	$2(2k - 3)x - k = 3 \Rightarrow x = \frac{k + 3}{4k - 6} \text{ or } k = \frac{3 + 6x}{4x - 1}$	DM1	Differentiating and for x or k
	Either $(2k - 3)\left(\frac{k + 3}{4k - 6}\right)^2 - k\left(\frac{k + 3}{4k - 6}\right) - (k - 2) = 3\left(\frac{k + 3}{4k - 6}\right) - 4$ Or $4x\left(\frac{3x^2 + 3x - 6}{2x^2 - x - 1}\right) - 6x - \left(\frac{3x^2 + 3x - 6}{2x^2 - x - 1}\right) = 3$	DM1	Substituting the equation or the $k = \frac{3x^2 + 3x - 6}{2x^2 - x - 1}$ into derivative (dependent on previous M mark)
	$9k^2 - 54k + 81 [=0]$ [leading to $k^2 - 6k + 9 = 0$]	M1	Simplifying and their 3-term quadratic (or solving for x)
	$k = 3$	A1	
			SC If M0, B1 for differentiating, 3 and solving for
		5	

NO. 2

Functions f and g are defined by

$$f : x \mapsto 3x - 2, \quad x \in \mathbb{R},$$

$$g : x \mapsto \frac{2x + 3}{x - 1}, \quad x \in \mathbb{R}, x \neq 1.$$

- i. Obtain expressions for $f^{-1}(x)$ and $g^{-1}(x)$, stating the value of x for which $g^{-1}(x)$ is not defined. [4]
- ii. Solve the equation $fg(x) = \frac{7}{3}$. [3]

Mark scheme

Question	Answer	Marks	Guidance
7.i	$(f^{-1}(x)) = \frac{x+2}{3}$ oe	B1	
	$y = \frac{2x+3}{x-1} \rightarrow (x-1)y = 2x+3 \rightarrow x(y-2) = y+3$	M1	Correct method to obtain $x =$, (or $y =$, if interchanged) but condone + / - sign errors
	$(g^{-1}(x) \text{ or } y) = \frac{x+3}{x-2}$ oe $\left(eg \frac{5}{x-2} + 1 \right)$	A1	Must be in terms of x
	$x \neq 2$ only	B1	FT for value of x from their denominator = 0
		4	
ii	$(fg(x)) = \frac{3(2x+3)}{x-1} - 2 (= \frac{7}{3})$	B1	Correct method from their $fg = \frac{7}{3}$ leading to a linear equation and collect like terms. Condone bomission of $2(x-1)$.
	$18x + 27 = 13x - 13$ or $3(4x + 11) = 7(x - 1)$ $(5x = -40)$	M1	
	Alternative method for question 7(ii)		
	$(f^{-1}(\frac{7}{3})) = \frac{13}{9}$	B1	
	$\frac{2x+3}{x-1} = \frac{13}{9} \rightarrow 9(2x+3) = 13(x-1) (\rightarrow 5x = -40)$	M1	Correct method from $g(x) =$ their $\frac{13}{9}$ leading to a linear

			equation and collect like terms.
	$x = -8$	A1	
		3	

NO. 3

Two points A and B have coordinates $(3a, -a)$ and $(-a, 2a)$ respectively, where a is a positive constant.

i. Find the equation of the line through the origin parallel to AB . [2]

ii. The length of the line AB is $3\frac{1}{3}$ units. Find the value of a . [3]

Mark scheme

i	Gradient of $AB = -3/4$	B1	Accept $-3a/4a$
	$y = -\frac{3}{4}x$ oe	B1FT	Answer must not include a . Ft on <i>their</i> <u>numerical</u> gradient
		2	
ii	$(4a)^2 + (3a)^2 = (10/3)^2$ soi	M1	May be unsimplified
	$25a^2 = 100 / 9$ oe	A1	
	$a = 2/3$	A1	
		3	

NO. 4

i. Prove the identity $\left(\frac{1}{\sin \theta} - \frac{1}{\tan \theta}\right)^2 \equiv \frac{1 - \cos \theta}{1 + \cos \theta}$. [3]

ii. Hence solve the equation $\left(\frac{1}{\sin \theta} - \frac{1}{\tan \theta}\right)^2 = \frac{2}{5}$, for $0^\circ \leq \theta \leq 360^\circ$. [4]

Mark scheme

i	$\left(\frac{1}{\sin \theta} - \frac{1}{\tan \theta}\right)^2 \equiv \frac{1 - \cos \theta}{1 + \cos \theta}$ $\left(\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta}\right)^2 = \frac{(1 - \cos \theta)^2}{\sin^2 \theta}$ $= \frac{(1 - \cos \theta)(1 - \cos \theta)}{1 - \cos^2 \theta} = \frac{1 - \cos \theta}{1 + \cos \theta}$	M1 M1 A1	3	Use of $\tan = \sin/\cos$ Use of $\sin^2 + \cos^2 = 1$. All correct. (NB ag. – ensure cancelling has been done)
ii	$\left(\frac{1}{\sin \theta} - \frac{1}{\tan \theta}\right)^2 = \frac{2}{5}$ $\frac{1 - \cos \theta}{1 + \cos \theta} = \frac{2}{5}$ $\cos \theta = \frac{3}{7}$ $\theta = 64.6^\circ \text{ or } 295.4^\circ$	M1 A1 A1 A1 ✓	4	Uses part (i) to obtain an eqn in $\cos \theta$ co co. ✓ for 360 – “1st answer”.

NO. 5

The coefficients of x and x^2 in the expansion of $(2 + ax)^7$ are equal. Find the value of the non-zero constant a . [3]

Mark scheme

Question	Answer	Marks	Guidance
1	$7C1 \times 2^6 \times a(x), 7C2 \times 2^5 \times [a(x)]^2$	B1 B1	SOI Can be part of expansion. followed by a^2 . ALT $2^7 [1 + ax/2]^7 \rightarrow 7C1 [a(x$
	$a = \frac{7 \times 2^6}{21 \times 2^5} = \frac{2}{3}$	B1	Ignore extra soln $a = 0$. Allow a allow an extra x in the answer
	Total:	3	

NO. 6

Relative to the origin O , the points A , B and C have position vectors given by

$$\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 3 \\ -2 \\ -4 \end{pmatrix}.$$

The quadrilateral $ABCD$ is a parallelogram.

a. Find the position vector of D . [3]

b. The angle between BA and BC is θ .

Find the exact value of $\cos \theta$. [3]

c. Hence find the area of $ABCD$, giving your answer in the form $p\sqrt{q}$, where p and q are integers. [4]

Mark scheme

Question	Answer	Marks	Guidance
a	Obtain a vector for one side of the parallelogram	B1	e.g. $\overrightarrow{AB} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ or $\overrightarrow{BC} = \begin{pmatrix} -1 \\ -5 \\ -6 \end{pmatrix}$.
	Correct method to obtain $\pm \overrightarrow{OD}$	M1	e.g. $\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{BC}$. MO if use $\overrightarrow{AB} = \overrightarrow{CD}$ or $\overrightarrow{BC} = \overrightarrow{DA}$.
	Obtain $\overrightarrow{OD} = \mathbf{i} - 4\mathbf{j} - 3\mathbf{k}$	A1	Any equivalent form. Accept coordinates.
		3	
b	Using the correct process, evaluate the scalar product $\overrightarrow{BA} \cdot \overrightarrow{BC}$	M1	(2+10–6) Scalar product of two relevant vectors. OE
	Using the correct process for the moduli, divide the scalar product by the product of the moduli.	M1	$\frac{2+10-6}{\sqrt{9} \times \sqrt{62}}$.
	Obtain answer $\frac{2}{\sqrt{62}}$	A1	ISW Or simplified equivalent i.e. $\frac{\sqrt{62}}{31}$
		3	
c	State or imply $\sin \theta = \sqrt{\frac{58}{62}}$	B1 FT	
	Use correct method to find the area of $ABCD$	M1	e.g. $2 \times \frac{1}{2} BA \times BC \sin \theta$. Condone decimals.
	Correct unsimplified expression for the area	A1 FT	e.g. $2 \times \frac{1}{2} \times 3 \times \sqrt{62} \times \sin \theta$. Condone decimals. Follow their sides and angle.
	Obtain answer $3\sqrt{58}$	A1	Correct only.
		4	

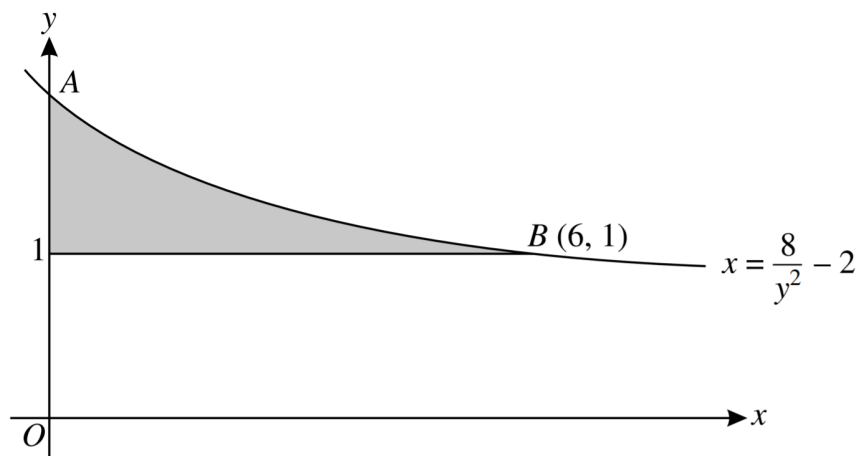
NO. 7

A curve has equation $y = 3x^3 - 6x^2 + 4x + 2$. Show that the gradient of the curve is never negative. [3]

Mark scheme

2	$\frac{\partial y}{\partial x} = 9x^2 - 12x + 4$	M1A1	3	
	$(3x - 2)^2 \geq 0$	A1		

NO. 8



The diagram shows part of the curve $x = \frac{8}{y^2} - 2$, crossing the y -axis at the point A. The point $B(6, 1)$ lies on the curve. The shaded region is bounded by the curve, the y -axis and the line $y = 1$. Find the exact volume obtained when this shaded region is rotated through 360° about the **y -axis**. [6]

Mark scheme

5	$x = \frac{8}{y^2} - 2 ; \text{ at } x = 0, y = 2$ $\rightarrow x^2 = \frac{64}{y^4} - \frac{32}{y^2} + 4$ $\text{Integral of } x^2 = \frac{64y^{-3}}{-3} - \frac{32y^{-1}}{-1} + 4y$ $\text{Uses limits 1 to 2}$ $\rightarrow 6\frac{2}{3}\pi$	B1 B1B1B1 M1 A1	6	co All co. Uses 1 to 2 or 2 to 1. co.
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NO. 9

Marco has four boxes labelled K , L , M and N . He places them in a straight line in the order K , L , M , N with K on the left. Marco also has four coloured marbles: one is red, one is green, one is white and one is yellow. He places a single marble in each box, at random. Events A and B are defined as follows.

A : The white marble is in either box L or box M .

B : The red marble is to the left of both the green marble and the yellow marble.

Determine whether or not events A and B are independent. [3]

Mark scheme

Question	Answer	Marks	Guidance
5	$P(A) = \frac{1}{2}, \quad P(B) = \frac{8}{24} = \frac{1}{3},$	B1	Both stated, accept unsimplified.
	$P(A \cap B) = \frac{1}{6}$	M1	Evidence that independence properties not used.
	$P(A) \times P(B) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ so events are independent	A1	Evaluated and conclusion stated. $P(A) \times P(B)$ and $P(A \cap B)$ seen.
		3	

NO. 10

Lengths of a certain type of white radish are normally distributed with mean μ cm and standard deviation σ cm. 4% of these radishes are longer than 12 cm and 32% are longer than 9 cm. Find μ and σ . [5]

Mark scheme

2	$1.751 = \frac{12 - \mu}{\sigma}$	B1	Rounding to ± 1.75 seen
	$0.468 = \frac{9 - \mu}{\sigma}$	B1	± 0.468 seen
	$\sigma = 2.34$	M1	An eqn with a z-value, μ and σ no $\sqrt{\sigma}$, no σ^2
	$\mu = 7.91$	M1	Sensible attempt to eliminate μ or σ by substitution or subtraction, need a value
		A1 5	correct answers